

todo-report

Purpose

`todo-report` is a small Go CLI for inspecting TODO health across local Git repositories.

It answers three practical questions:

1. How old is the open work?
2. How do TODOs differ across branches?
3. Is the TODO structure valid?

The tool is intentionally small, local-first, and file-oriented.

It does not track promises.

It does not record assessments.

It does not replace `commitment-ledger`.

It produces useful reports that can later become evidence inputs to `commitment-ledger`.

Relationship to PromiseGrid

PromiseGrid emphasizes durable records, explicit evidence, local interpretation, branch/context awareness, and avoiding hidden global truth.

`todo-report` follows that style by:

- observing local repos without editing them
- treating branch state as local and specific
- reporting evidence rather than making promises
- making TODO structure visible and auditable
- producing machine-readable output for later tools

This is a utility companion to `commitment-ledger`.

Language

Go.

Reasons:

- Small compiled CLI
 - Easy local Git and file inspection
 - Fits the existing team tooling style
 - Can share parsing ideas with `commitment-ledger`
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Core Commands

1. Age Report

```
todo-report age --repo /path/to/repo --branch main
```

Shows open TODOs ordered by age.

Example output:

TODO-binap	142 days	TODO/TODO.md
TODO-ravud	91 days	TODO/TODO-ravud-demo.md
003	14 days	TODO/TODO.md

Age should be derived from Git history where possible.

Preferred rule:

```
Age = first commit date where this TODO ID appeared.
```

Fallback:

```
Age = first observed date in local cache.
```

2. Drift Report

```
todo-report drift --repo /path/to/repo --branch-a main --branch-b jj
```

Compares TODO state across two branches.

Example output:

```
Only in main:
  TODO-binap

Only in jj:
  TODO-fodam

Completed in jj only:
  TODO-ravud

Completed in main only:
  003
```

Branch identity matters.

A checked TODO on `jj` does not mean the TODO is checked on `main`.

3. Lint Report

```
todo-report lint --repo /path/to/repo --branch main
```

Checks TODO structure.

Detect:

- duplicate TODO IDs
- missing detail files
- malformed proquint TODO IDs
- malformed numeric TODO IDs
- invalid checkbox syntax
- referenced subtask not found
- duplicate subtask IDs
- checked parent with unchecked required subtasks
- broken backtick detail-file links

Example output:

```
ERROR duplicate_id TODO-binap TODO/TODO.md:12
ERROR missing_detail_file TODO-ravud TODO/TODO-ravud-demo.md
WARNING checked_parent_open_subtask TODO-hokos/2.1
```

4. Health Report

```
todo-report health --repo /path/to/repo --branch main
```

Combines age, drift summary, and lint summary.

Example output:

```
Repo: wire-lab  
Branch: main  
  
Open TODOs: 37  
Completed TODOs: 18  
Oldest open TODO: TODO-binap, 142 days  
  
Lint errors: 2  
Lint warnings: 7
```

For multi-branch health:

```
todo-report health --repo /path/to/repo --branch main --compare jj
```

Example:

```
Branch drift items: 4  
Completed only on jj: 2  
Only on main: 1  
Only on jj: 1
```

Supported TODO Formats

Numeric Format

```
001 - Add route support  
002 - Add malformed packet tests
```

Proquint Format

- [] TODO-binap - Lock and flesh out README outline (`TODO/TODO-binap-readme-outline-lock.md`)
- [x] TODO-ravud - Ship welcome flow (`TODO/TODO-ravud-ship-welcome-flow.md`)

Detail File Subtasks

- [] 1. Add route
- [] 2. Add tests
- [] 2.1 Add malformed input test
- [x] 2.2 Add happy path test

Subtask target format:

TODO-binap/2.1

Data Model

TodoItem

```
{
  "repo": "wire-lab",
  "branch": "main",
  "commit_hash": "abc123",
  "todo_id": "TODO-binap",
  "title": "Lock and flesh out README outline",
  "status": "open",
  "source_file": "TODO/TODO.md",
  "detail_file": "TODO/TODO-binap-readme-outline-lock.md"
}
```

Subtask

```
{
  "repo": "wire-lab",
  "branch": "main",
  "commit_hash": "abc123",
```

```
"parent_id": "TODO-binap",
"subtask_id": "2.1",
"target": "TODO-binap/2.1",
"title": "Add malformed input test",
"status": "open",
"source_file": "TODO/TODO-binap-readme-outline-lock.md"
}
```

LintFinding

```
{
  "severity": "error",
  "code": "missing_detail_file",
  "todo_id": "TODO-ravud",
  "file": "TODO/TODO.md",
  "line": 12,
  "message": "Detail file does not exist."
}
```

Output Formats

Default output is human-readable text.

Support JSON for tool chaining:

```
todo-report lint --repo /path/to/repo --json
todo-report age --repo /path/to/repo --json
todo-report drift --repo /path/to/repo --branch-a main --branch-b jj --json
```

JSON output makes it possible for `commitment-ledger` to import reports later as evidence.

Architecture

Suggested Go layout:

```
todo-report/
cmd/
  todo-report/
    main.go
```

```
internal/  
  gitrepo/  
    gitrepo.go  
    history.go  
  todo/  
    parser.go  
    detail_parser.go  
    model.go  
  age/  
    age.go  
  drift/  
    drift.go  
  lint/  
    lint.go  
  report/  
    text.go  
    json.go  
README.md
```

Package Responsibilities

gitrepo

- verify local Git repo
- get branch
- get commit hash
- inspect file contents on branches
- find first commit where TODO ID appeared

todo

- parse `TODO/TODO.md`
- parse detail files
- detect numeric IDs
- detect proquint IDs
- detect checked state
- produce structured TODO records

age

- compute TODO age
- sort oldest first

drift

- compare TODOs across branches
- compare checked state

- compare detail-file presence
- compare subtasks

lint

- validate TODO structure
- emit findings

report

- render text and JSON

Integration with commitment-ledger

`todo-report` should remain independent.

Future integration path:

```
todo-report JSON output
      |
      v
commitment-ledger evidence import
```

Possible future command:

```
commitment-ledger evidence import --from todo-report.json
```

Examples:

- lint report becomes evidence of malformed TODO structure
- drift report becomes evidence of branch-local divergence
- age report becomes evidence of stale available work

Non-Goals

Do not implement:

- promise creation
- commitment assessment
- CAS storage
- signed grid artifacts

- pCID protocol docs
- GitHub API
- Gitea API
- source repo editing
- web UI

Those belong to `commitment-ledger` or future tools.

Acceptance Criteria

v0.1 is complete when:

1. `todo-report age` works on a local Git repo.
 2. `todo-report drift` compares two branches.
 3. `todo-report lint` catches duplicate IDs, missing detail files, and malformed TODOs.
 4. `todo-report health` summarizes repo TODO health.
 5. Text output is readable.
 6. JSON output is available.
 7. The tool never modifies source repos.
 8. Tests cover numeric TODOs, proquint TODOs, detail-file subtasks, drift, age, and lint findings.
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Summary

`todo-report` is a small PromiseGrid-adjacent CLI.

It observes TODO work across Git branches and reports age, drift, and structure quality.

It is useful by itself.

Later, its reports can become evidence for `commitment-ledger`.